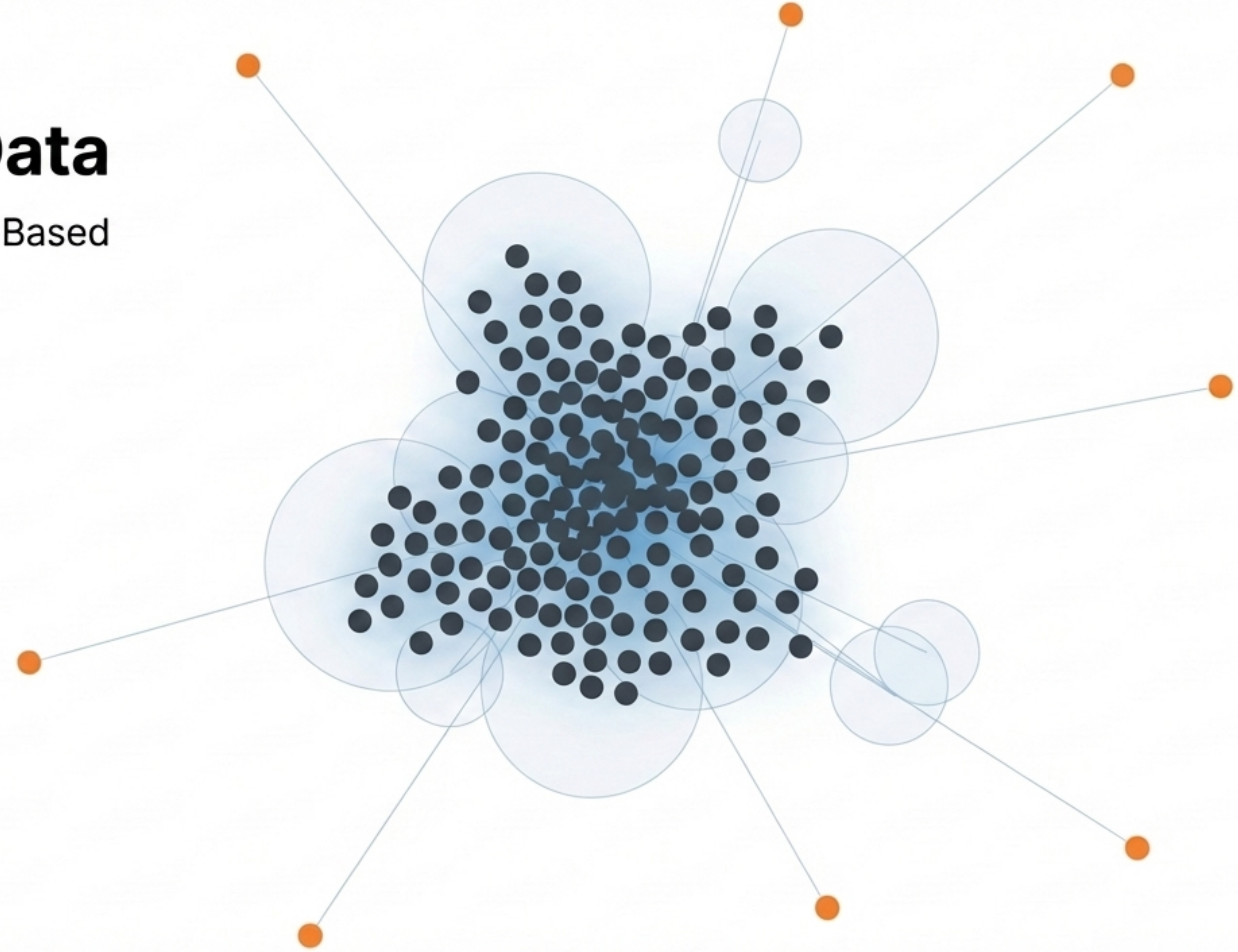
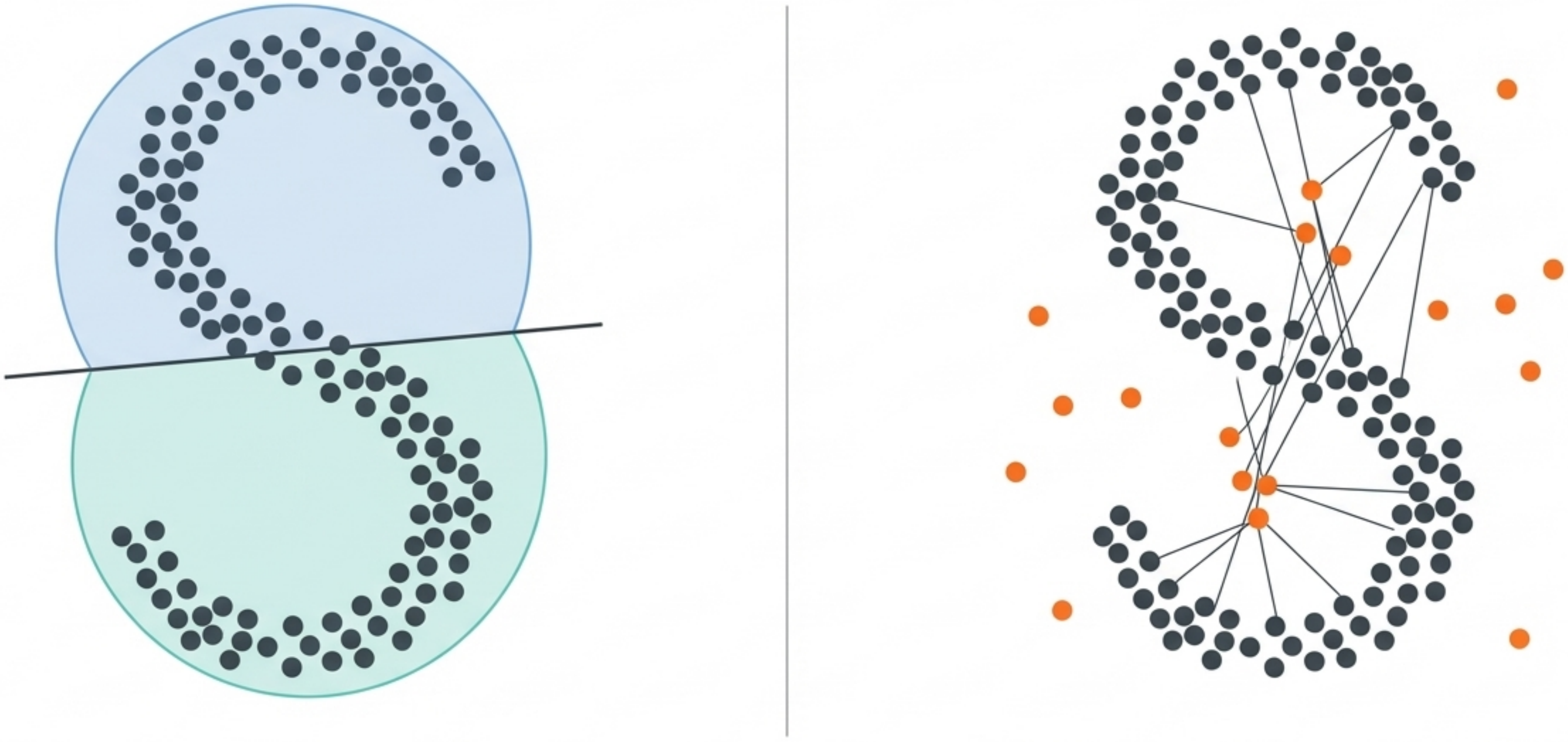
Mapping the Geometry of Data

An Introduction to Density-Based Clustering and DBSCAN



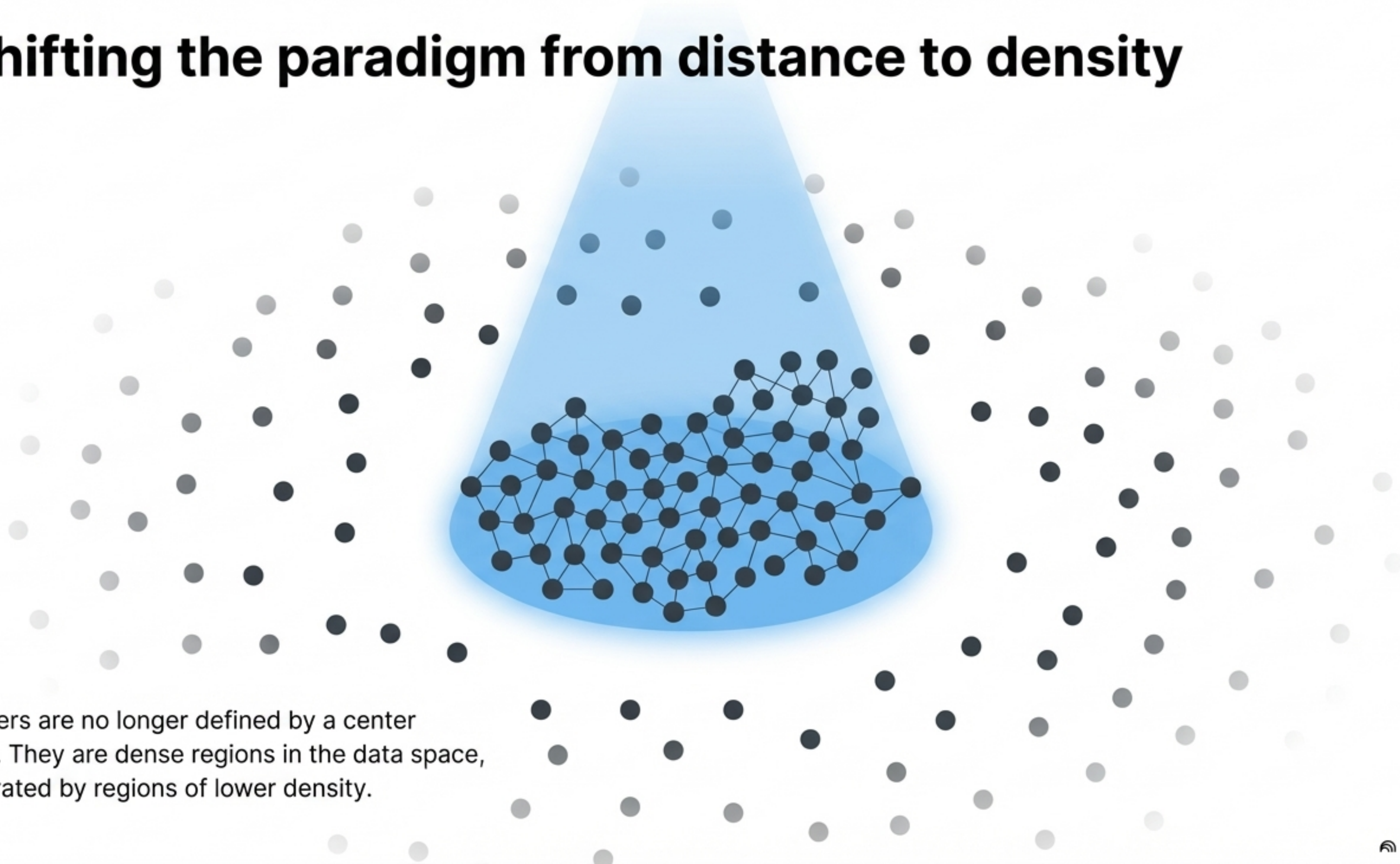
The spherical trap of traditional clustering methods



→ Forces spherical partitions

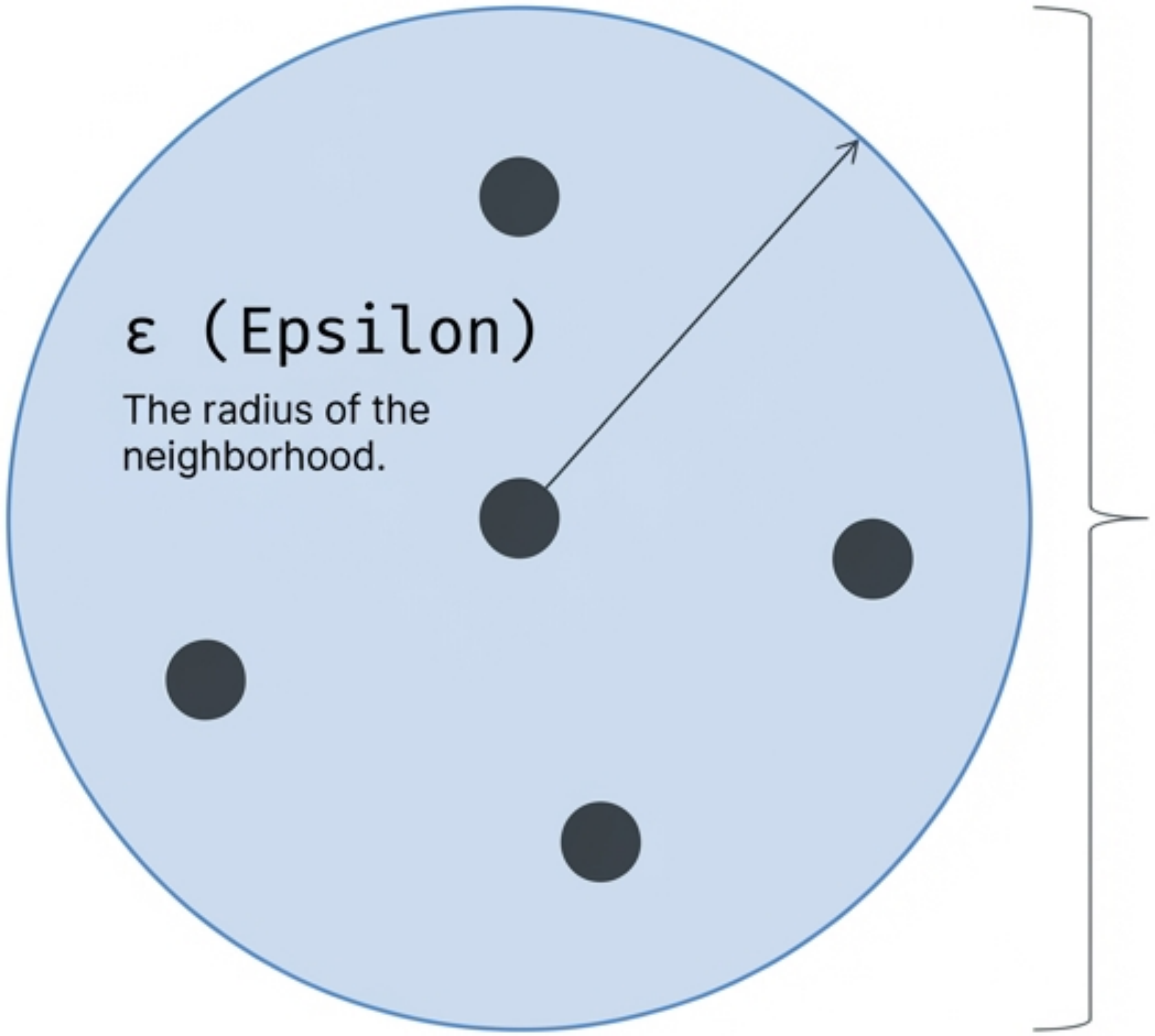
→ Highly sensitive to noise bridges

Shifting the paradigm from distance to density



Clusters are no longer defined by a center point. They are dense regions in the data space, separated by regions of lower density.

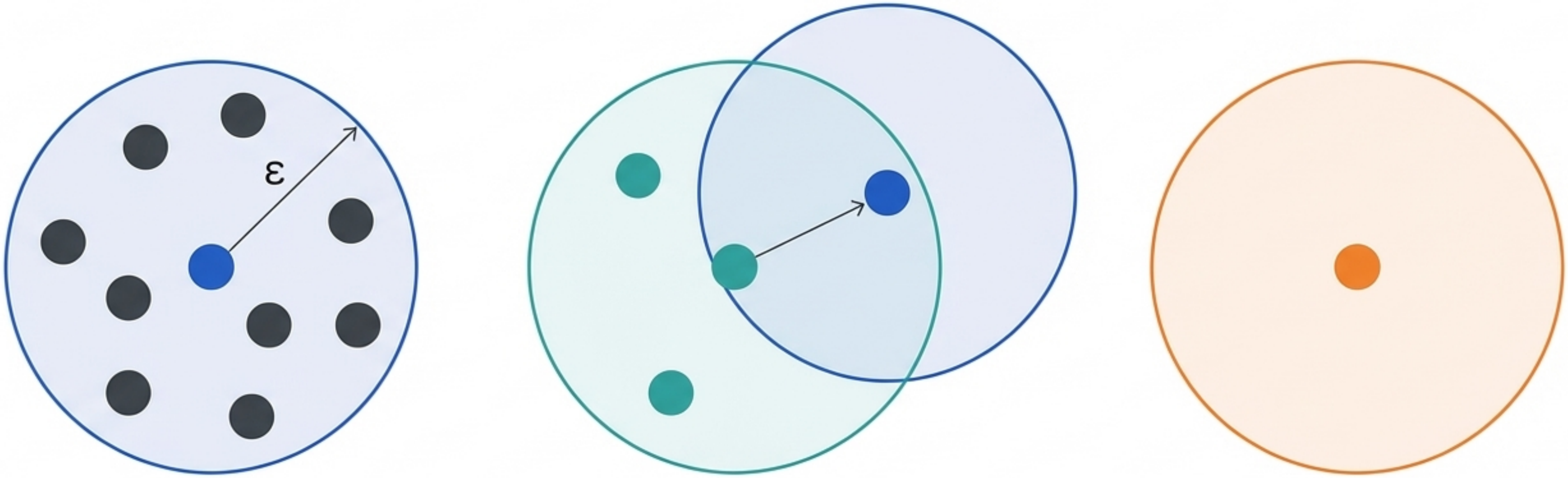
The two geometric rules of density



MinPts

The density threshold: the minimum number of points required inside the ε-neighborhood.

The anatomy of a data space



Core Point: ε-neighborhood contains ≥ `MinPts`.

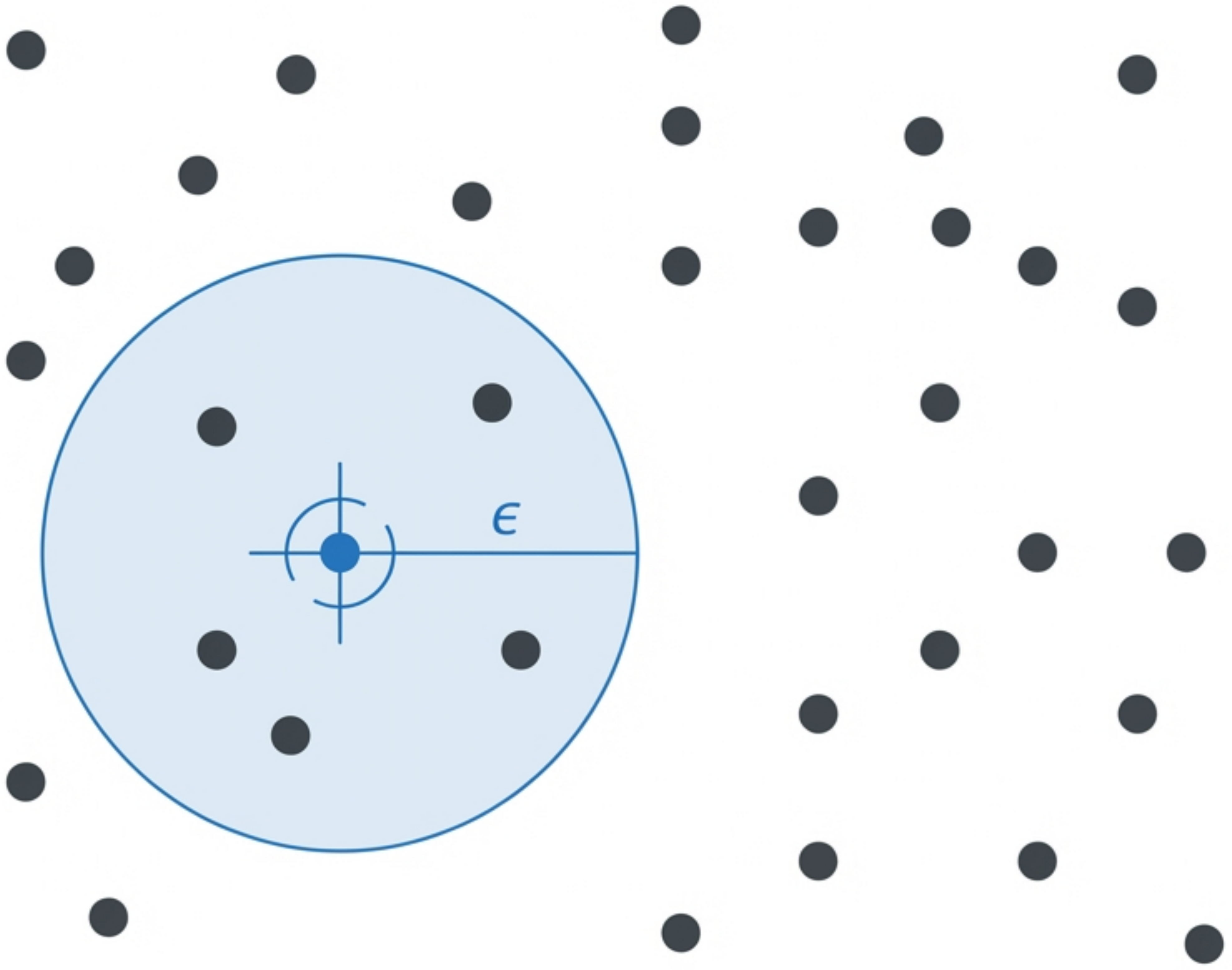
Border Point: < `MinPts`, but falls within a Core point's radius.

Noise / Outlier: Isolated and unreachable.

Running DBSCAN: The Seed Phase

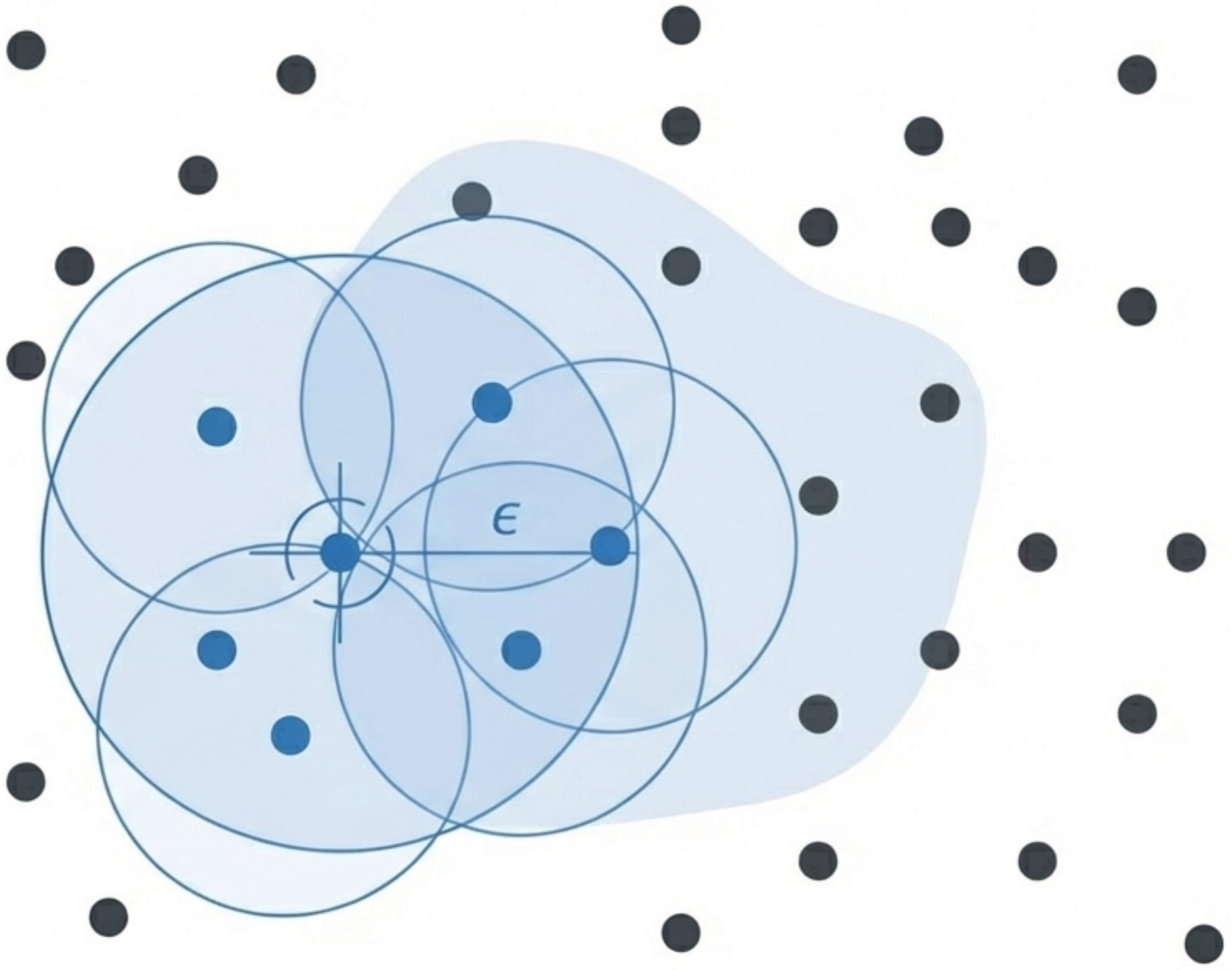
1. Randomly select an unvisited point.

The point passes the `MinPts` test. It becomes a Core “**Seed**” and a new cluster is born.



Running DBSCAN: The Expansion Phase

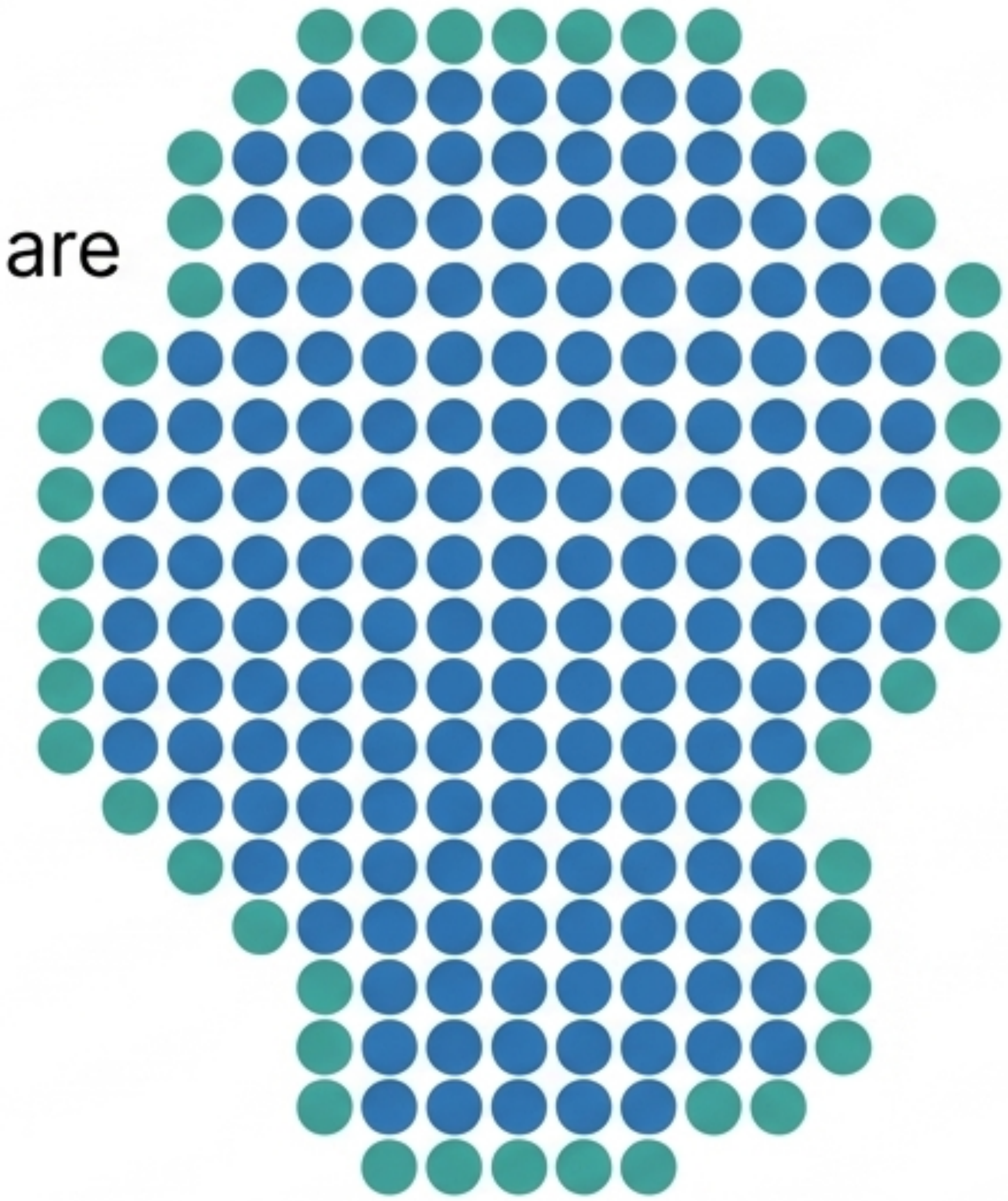
2. Evaluate neighbors. If they are also Core points, add their neighborhoods to the candidate set. The cluster expands organically.

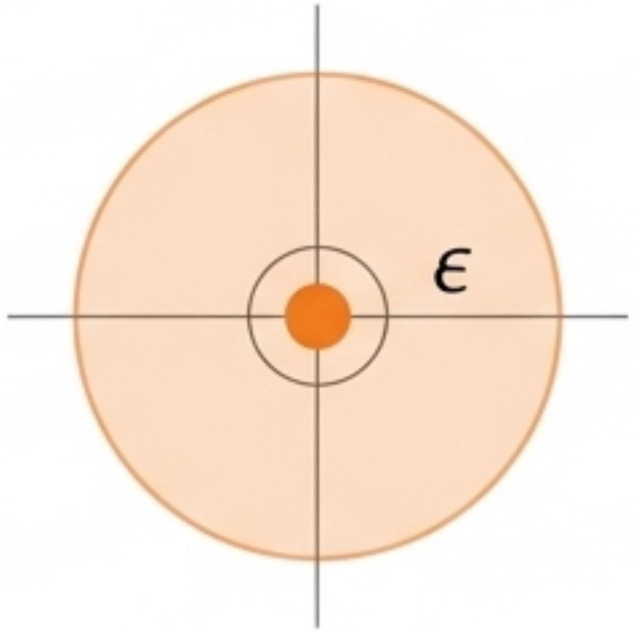


Running DBSCAN: Termination and Noise

3. Expansion stops at Border points. Unvisited points that fail the density test are flagged as Noise.

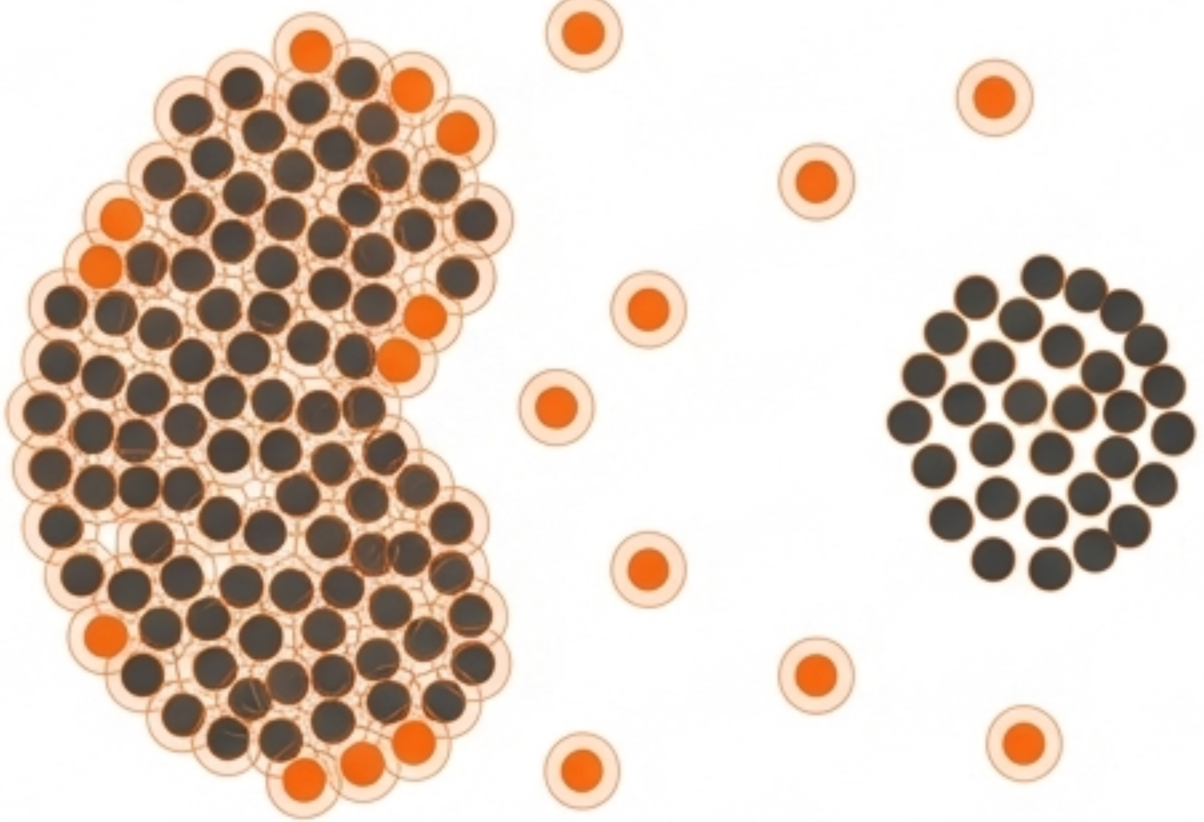
Repeat until all points are visited.





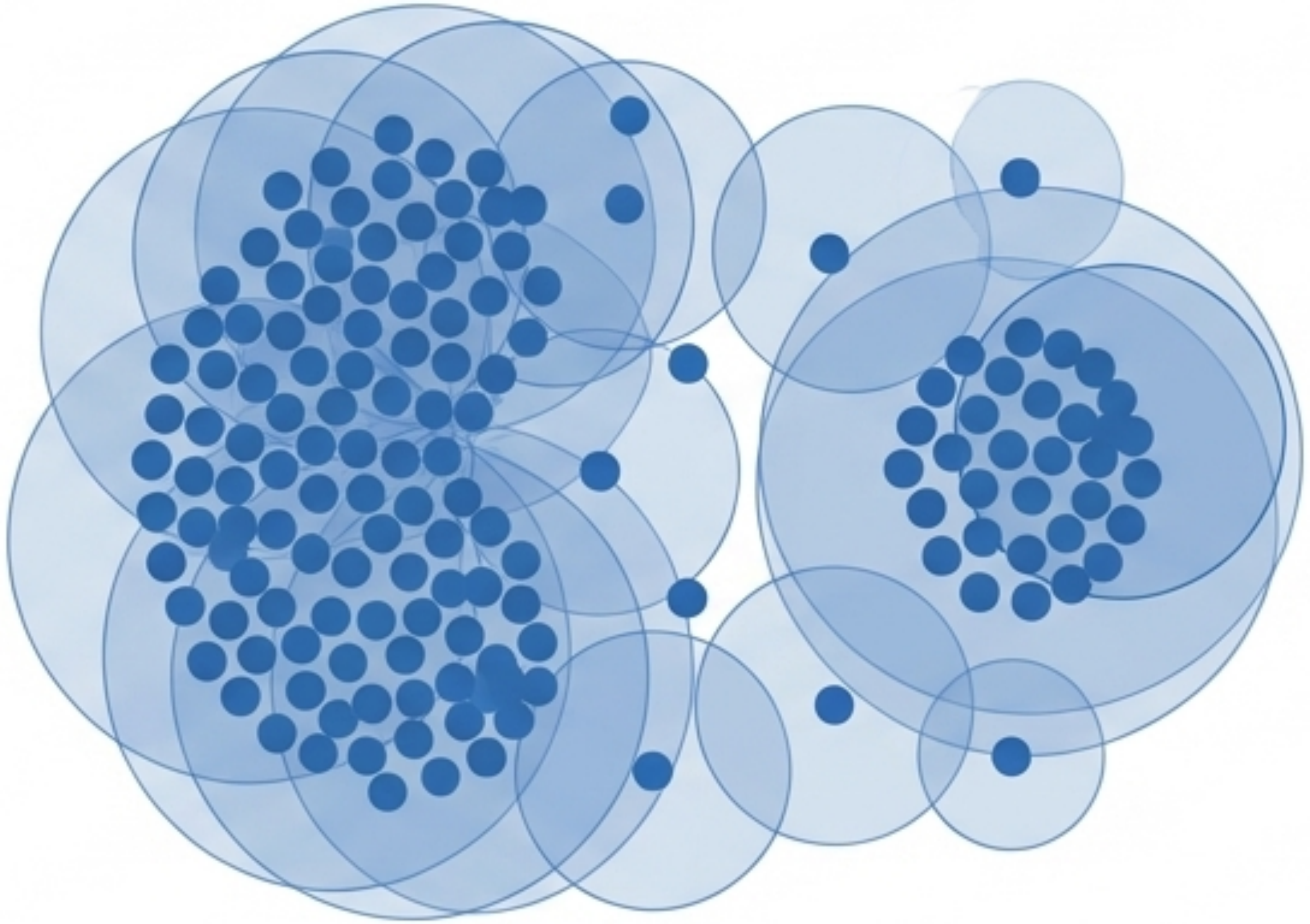
The calibration problem: Finding the perfect radius

ε is too small.



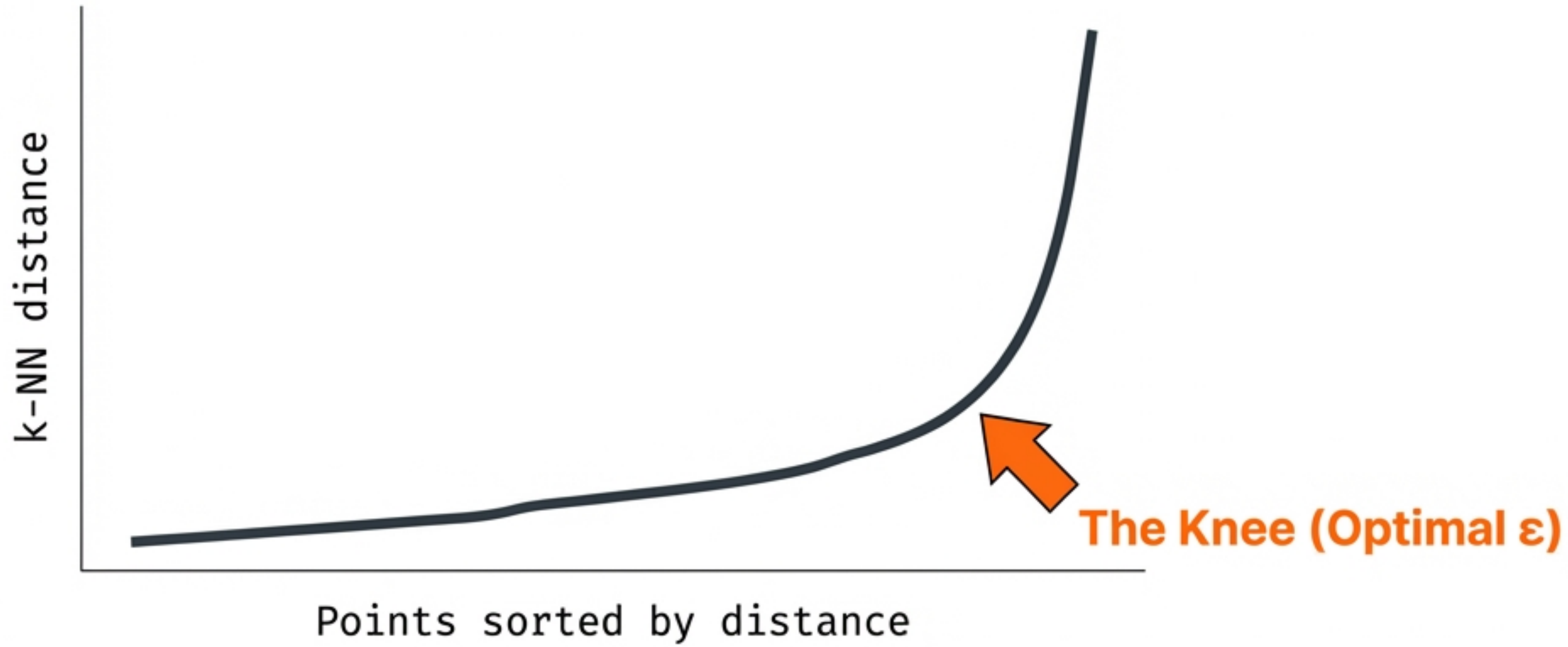
Sparser clusters are incorrectly defined as noise.

ε is too large.



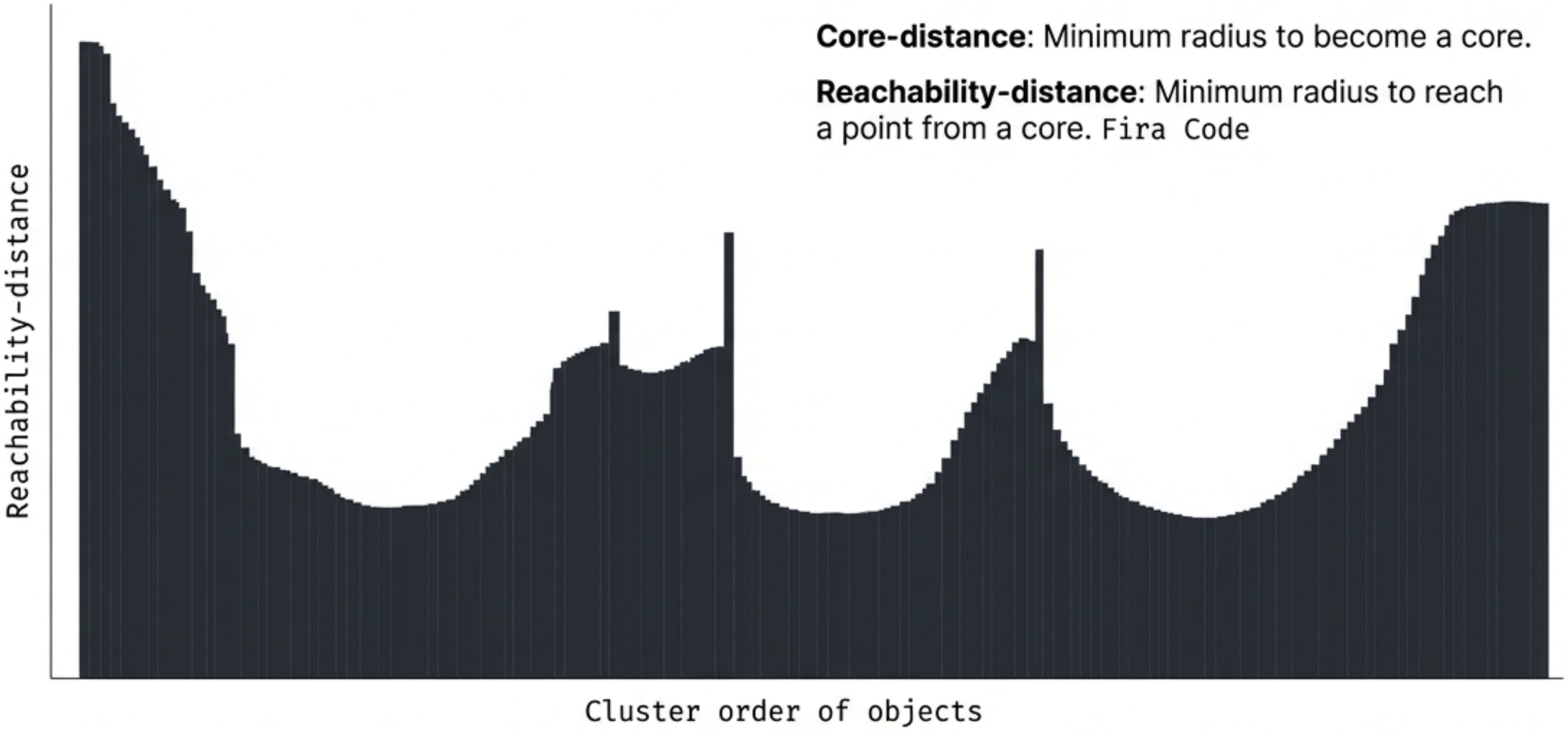
Distinct, dense clusters are incorrectly merged together.

Finding the 'Knee': The k-distance graph



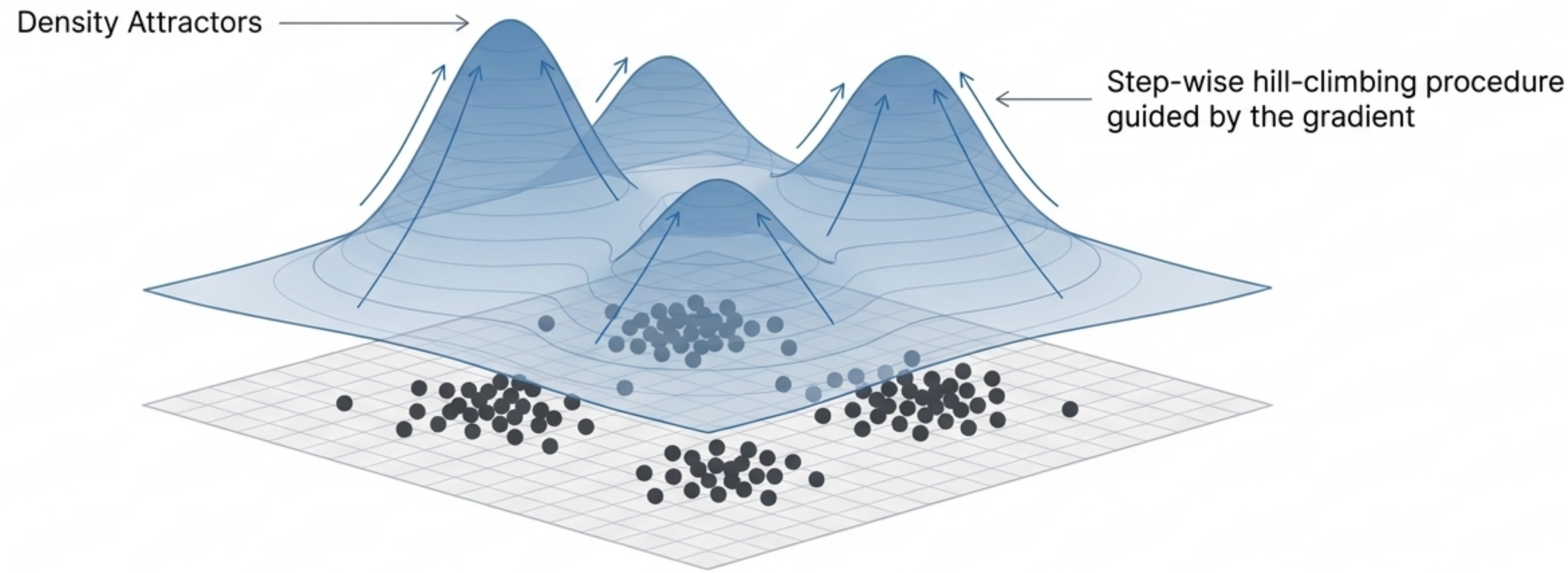
Plot the distance to the k-th nearest neighbor (k = MinPts). The optimal ε is located at the threshold where the curve shows a strong upward bend.

Beyond DBSCAN: Handling varying densities with OPTICS



OPTICS does not explicitly produce a hard clustering. Instead, it extracts a hierarchical clustering ordering, allowing you to identify clusters at any density threshold.

Beyond DBSCAN: Continuous distributions with DENCLUE



Instead of radius counting, DENCLUE uses a Gaussian kernel to estimate a continuous density function. Points 'climb' the gradient to assign themselves to a density peak.

The clustering synthesis matrix

	K-Means	DBSCAN	OPTICS	DENCLUE
Arbitrary Shapes?	No (Spherical only)	Yes	Yes	Yes
Handles Noise/Outliers?	No (Highly sensitive)	Yes	Yes	Yes (Invariant against noise)
Primary Input Parameters	k (number of clusters)	`ε, MinPts`	`MinPts`	Smoothing parameter (h), Noise threshold (ξ)
Handles Varying Densities?	N/A	No (Uses global threshold)	Yes (Outputs cluster ordering)	Yes